

# AUTHENTICLENS

## KNOWLEDGE BASE - DOCUMENT 1 OF 9

# DOCUMENT 1 OF 9 - THE AI ARTIFACT LEXICON

Corrected forensic reference for AI-image flaws, false positives, and evidence weighting

Version 2.0 - April 2026. Fully revised and corrected from the original v1.0 document. This version integrates the audit findings, errata, Lightroom/editorial separation, mobile/compression caveats, provenance limits, and Gemini / Nano Banana / Nano Banana Pro execution requirements.

Prepared for: AuthenticLens - The Realism Architect

Target use: Gemini Gem knowledge-base upload

Primary image workflow: Gemini Pro reasoning + Nano Banana / Nano Banana Pro image generation and editing

Language: English (Canada)

## Purpose

This document is the corrected artifact lexicon for AuthenticLens. It catalogues visual failure modes associated with AI-generated images while preventing the common false positive: confusing AI-generation artifacts with heavy Lightroom/editorial processing, phone-camera computation, low-resolution uploads, or social-media compression.

## Intended Gemini Behaviour

- Use this lexicon as a red-flag vocabulary, not as a standalone verdict engine.
- Never classify an image as AI-generated based only on smooth skin, high contrast, a vignette, colour grading, sharpened texture, or a cinematic style.
- Prioritise structural failures: anatomy topology, text fidelity, shadow logic, reflection logic, impossible geometry, object boundaries, and scene plausibility.
- When evidence is mixed or the file is low-resolution, compressed, or screenshot-based, state uncertainty.
- If the user says the photo is real, recalibrate: explain which cues are likely editing artifacts rather than AI-generation artifacts.

## Version 2.0 Corrections Applied

| Old weakness                                                   | Updated correction                                                                                                                                                    |
|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hands treated as a near-decisive AI signal.                    | Hands remain high-weight evidence only when topology fails. Motion blur, low light, noise reduction, compression, or shallow depth of field can distort real hands.   |
| Single AI confidence score encouraged binary overclaiming.     | Use this lexicon with Document 4 four-axis scoring: Authenticity, Heavy Edit, AI Generation, Composite/Manipulation, plus Uncertainty.                                |
| AI glow overlapped with editorial bloom and Lightroom effects. | AI glow only raises AI confidence when paired with structural/physics failures. Alone, it may indicate heavy editing.                                                 |
| No phone/compression caveat.                                   | Every artifact must be tested against phone processing, JPEG/social compression, screenshots, upscaling, and AI-denoise workflows.                                    |
| Typography and logo failures were treated broadly.             | Readable but compressed text can be real; morphing glyphs, inconsistent baselines, impossible ligatures, and unstable repeated characters are stronger AI indicators. |

## Core Evidence Principle

Aesthetic polish is weak evidence. Structural contradiction is strong evidence. Converging structural failures across independent categories are the basis for AI confidence.

## Category 1 - Anatomy and Human Structure Artifacts

Human anatomy remains a critical diagnostic area, but modern image models can produce plausible first-pass anatomy. AuthenticLens must inspect topology, joint logic, object interaction, and regional consistency rather than simply looking for obvious extra fingers.

| Subcategory                 | AI-generation artifact                                                                                                                           | False-positive caveat                                                                                                        | Confidence weight                                           |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Hands and fingers           | Extra or missing digits, fused joints, impossible thumb origin, fingers bending without knuckles, nails on wrong surfaces, palm topology errors. | Real motion blur, low-light noise reduction, compression, and occlusion can smear fingers. Do not overclaim from blur alone. | High when topology is impossible; medium when only blurred. |
| Wrists and limb connections | Hand does not connect structurally to wrist; forearm angle contradicts palm orientation; limb passes through object.                             | Wide-angle distortion and pose foreshortening can look unusual.                                                              | High if bone path is impossible.                            |
| Eyes and catchlights        | Pupils different sizes without medical/lighting cause; catchlights contradict light source; iris texture collapses into generic radial pattern.  | Beauty retouching and compression can simplify iris detail.                                                                  | Medium to high with lighting contradiction.                 |
| Teeth and mouth             | Too many teeth, fused teeth, gumline absent, tongue too smooth or shapeless, smile geometry impossible.                                          | Whitening, retouching, blur, and compression can reduce gum detail.                                                          | High when count/structure fails.                            |
| Ears                        | Helix/antihelix fused, tragus missing, ear attached impossibly, both ears mirror-identical.                                                      | Hair or shallow focus can obscure ears.                                                                                      | Medium.                                                     |
| Hair and facial hair        | Hairline melts into scalp, beard disconnected from face, random strand direction, edge halo fused with background.                               | Backlight, masking, and subject cutouts create halos in real edits.                                                          | Medium unless paired with boundary failure.                 |
| Body proportions            | Torso/limb length shifts, shoulder asymmetry without pose cause, impossible posture or body-object intersection.                                 | Lens compression, pose, clothing, and perspective can exaggerate proportions.                                                | Medium to high.                                             |

Analyst instruction: Zoom to 200 percent when possible, but report resolution limitations. If the uploaded

image is small or heavily compressed, downgrade anatomy confidence unless the structural failure is obvious.

## Category 2 - Text, Typography, Logos, and Digital Displays

Text fidelity remains one of the strongest AI indicators, especially for street signs, product labels, luxury logos, screen interfaces, numbers, and small repeated type. Current models can render more readable text than older models, so the inspection must be glyph-level and context-aware.

| Artifact type          | High-confidence AI signs                                                                                      | Could also be editing/compression                                                       |
|------------------------|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Pseudo-text            | Characters resemble writing but do not form real words; glyphs morph together; repeated fake letter clusters. | Low-resolution signs can become unreadable after social compression.                    |
| Baseline drift         | Letters in the same word float up/down, rotate individually, or change height without design logic.           | Perspective on real angled signs can skew baselines coherently.                         |
| Font inconsistency     | Serif/sans-serif mixed inside one word; random stroke thickness; inconsistent counters in a,b,d,e,o,p,q.      | Compression can blur counters but does not invent random font systems.                  |
| Numbers and dates      | Impossible time displays, malformed digits, mixed numeral systems, addresses that change across signs.        | Glare/reflection can obscure digital displays.                                          |
| Logos and luxury marks | Known logos are approximate, misspelled, mirrored, or rendered as symbol-like pseudo-text.                    | Avoid overclaiming when a real logo is partially hidden, motion-blurred, or compressed. |
| Screen UI              | App layout contains impossible icons, inconsistent alignment, unreadable labels, or fake notification logic.  | Screens can bloom, moire, or clip in real photography.                                  |

Prompting note for edits: when text/logo fidelity is not essential, instruct Nano Banana / Nano Banana Pro to avoid generating new readable text. When fidelity matters, request minimal, simple, front-facing text and verify after generation.

## Category 3 - Repeating Patterns and Fine Texture Hallucinations

AI models often approximate pattern density locally but fail global repeat logic. This is visible in fabrics, brickwork, tiles, railings, window grids, chains, jewellery, woven materials, product stitching, and natural textures.

| Pattern domain      | AI artifact                                                                                     | Real-world check                                                                                                      |
|---------------------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Woven fabric        | Warp/weft pattern dissolves, threads merge, weave changes direction without seam or fold.       | Real cloth changes direction at folds, seams, and stretched areas; the pattern should still obey fabric construction. |
| Stitching and seams | Stitches stop midline, spacing changes randomly, seam does not follow garment edge.             | Handmade garments may have irregularity, but construction logic should remain.                                        |
| Architecture grids  | Windows, bricks, railings, tiles drift, scale, or repeat irregularly without perspective cause. | Lens distortion and perspective create coherent convergence, not random drift.                                        |
| Jewellery/chains    | Links overlap flatly rather than interlock; chain segments melt into skin or fabric.            | Reflections and shallow focus may obscure link structure but should not change topology.                              |
| Natural patterns    | Leaves, tree bark, grass, snow, water ripples repeat procedurally or melt into objects.         | Real organic patterns are irregular but physically constrained.                                                       |

## Category 4 - Object Boundaries, Edges, and Compositing Traces

| Boundary issue   | AI-generation sign                                                        | Composite/editing sign                                                    | Action                                                                                                          |
|------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Melted edges     | Object and background share texture or colour as if one surface.          | Cutout halo or mask fringe around subject.                                | Classify as AI if the object itself is structurally fused; classify as composite if cutout boundary is visible. |
| Hair/skin edges  | Hair dissolves into background without strand logic.                      | Selection mask halo, over-sharpened edge, or background-removal artifact. | Check if the face/body identity remains photographic but edge is edited.                                        |
| Contact edges    | Shoes, chair legs, hands, or objects float without contact shadows.       | Inserted subject may lack ambient occlusion.                              | Assess with Composite score and lighting consistency.                                                           |
| Occlusion errors | Objects pass through one another; foreground/background order impossible. | Poor manual composite can also produce it.                                | High manipulation evidence; classify based on local vs global failure.                                          |

## Category 5 - Light, Shadow, Reflection, and Scene Physics Failures

Lighting is stronger evidence than style. Real scenes can have complex mixed lighting, but they still obey direction, falloff, contact, reflection, and occlusion logic.

| Cue          | AI failure                               | Editorial false positive                 | Diagnostic rule                       |
|--------------|------------------------------------------|------------------------------------------|---------------------------------------|
| Cast shadows | Shadow direction changes between objects | Dodging/burning may lighten shadows; HDR | Look for geometry, not only darkness. |

|                     |                                                                                                                  |                                                                           |                                                                          |
|---------------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------|
|                     | without secondary source explanation; shadows detach or vanish.                                                  | may flatten them.                                                         |                                                                          |
| Contact shadows     | Feet, objects, or furniture float with no grounding.                                                             | Cutouts and background swaps may also miss contact shadows.               | Strong composite/manipulation cue when subject is real.                  |
| Reflections         | Mirror/window/water reflection shows different pose, missing object, wrong light direction, or impossible angle. | Dirty glass, curved surfaces, or polarizers can distort real reflections. | High AI/manipulation evidence if reflection contradicts object identity. |
| Specular highlights | Highlights appear decorative and unrelated to light source or material.                                          | Beauty retouching can add catchlights or shine.                           | Check consistency across eyes, metal, glass, skin, and fabric.           |
| Global illumination | No bounce light where surfaces should reflect colour; shadows have no environmental tint.                        | Overprocessed contrast can suppress bounce light.                         | Medium evidence; combine with other signs.                               |

## Category 6 - The AI Glow and Over-Perfect Aesthetic

AI glow is no longer enough for classification. It overlaps with real editorial effects such as soft glow, halation, mist filters, HDR tone mapping, skin smoothing, clarity/dehaze, highlight bloom, and social-media filters.

| Visual look      | Likely AI when...                                                                                    | Likely Lightroom/editorial when...                                                           |
|------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Plastic skin     | Skin lacks pores across all planes, ears/lips/neck share the same texture, lighting ignores anatomy. | Face is smoothed but hairline, neck, pores, clothing, and background remain physically real. |
| Perfect lighting | Every surface is evenly luminous without source/falloff logic.                                       | Studio retouching or HDR lifts shadows while preserving structure.                           |
| Hyper-saturation | Colours glow unnaturally while material physics fail.                                                | Colour grading shifts hue/saturation but objects remain coherent.                            |
| Halo/bloom       | Glow ignores bright-source position and material boundaries.                                         | Lens flare, mist filter, halation, backlight, or dehaze/clarity creates coherent glow.       |

Correction rule: if all suspicious cues are style-based and no structural failures are visible, classify as Heavy Edit rather than AI.

Category 7 - Material Physics Failures

| Material      | AI artifact signs                                                                                                          | Realism cues to expect                                                                                           |
|---------------|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Fabric        | Fabric behaves like rubber/plastic; folds ignore gravity; seams disappear; weave melts; logos warp with folds incorrectly. | Weight, drape, seam tension, compression folds, stitch spacing, lint, fibre direction, and realistic highlights. |
| Leather       | Texture repeats procedurally; pores too uniform; no crease logic at bend points.                                           | Creases, grain variation, edge wear, subtle specular changes.                                                    |
| Metal         | Reflections inconsistent; edges too liquid; tarnish/wear absent or painted randomly.                                       | Directional reflection, scratches, patina, edge wear, fingerprints, plausible highlight bands.                   |
| Glass         | Transparent object ignores refraction; reflections wrong; background behind glass unchanged.                               | Refraction, double reflections, glare, smudges, edge thickness, partial transparency.                            |
| Skin          | Uniform wax surface; no pores, fine lines, blood colour variation, beard shadow, under-eye texture.                        | Pores, varied texture, natural redness/brown undertones, micro shadows, hair follicles.                          |
| Road/pavement | Texture repeats or melts into sidewalk; road and curb share impossible plane.                                              | Curbs, drainage, cracks, road markings, surface wetness, snow/slush/weather consistency.                         |

Category 8 - Structural and Spatial Coherence Failures

- Vanishing points conflict without camera/lens explanation.
- Background scale contradicts subject scale.
- Sidewalk, curb, road, and building planes do not align physically.
- Furniture/object legs touch the ground at impossible angles.
- Repeated background objects mutate across the frame.
- Depth of field is applied decoratively rather than by distance from the focus plane.
- Lens compression does not match the apparent focal length or camera height.

Operational rule: perspective errors are strong evidence when they affect rigid structures such as buildings, roads, shelves, tables, screens, frames, and windows.

Category 9 - Model-Specific Visual Signatures: Updated Use

Model-specific signatures are less reliable over time because models update quickly. Treat them as weak-to-medium supporting cues unless confirmed by provenance, embedded labels, SynthID/watermark detection, or source disclosure.

| Signature type       | Examples                                                       | Use level                                 |
|----------------------|----------------------------------------------------------------|-------------------------------------------|
| Diffusion smoothness | Over-blended textures, painterly skin, soft object boundaries. | Medium only if structural failures exist. |

|                              |                                                                                    |                                              |
|------------------------------|------------------------------------------------------------------------------------|----------------------------------------------|
| Over-stylised editorial look | Perfect subject isolation, glossy contrast, symmetry, polished cinematic lighting. | Low by itself; could be real editorial work. |
| Text/logos                   | Improved but still inconsistent on small, dense, or perspective text.              | High when glyphs morph.                      |
| Hands/anatomy                | Gross failures less common; subtle topology failures remain.                       | High only when topology impossible.          |
| Gemini/Nano Banana output    | May follow complex edits well but can drift identity or over-clean texture.        | Use post-edit QC, not assumptions.           |

## Updated Confidence Weighting System

Document 1 no longer produces a single AI score by itself. It feeds evidence into Document 4. Use the following evidence weights before final classification.

| Evidence type                          | Weight        | Examples                                                                                                               |
|----------------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------|
| Very strong AI evidence                | 8-10          | Impossible anatomy topology; pseudo-text; impossible reflections; object intersections; global geometry contradiction. |
| Strong manipulation/composite evidence | 7-9           | Mismatched grain/noise between subject and background; edge halos; absent contact shadow; incompatible camera height.  |
| Medium AI/edit evidence                | 4-6           | Boundary softness, repeated texture, suspicious bokeh, excessive glow, uniform material texture.                       |
| Weak evidence / style only             | 1-3           | High contrast, vignette, colour grade, clarity/dehaze, smooth skin, HDR look, shallow DOF.                             |
| Uncertainty raisers                    | + uncertainty | Low resolution, screenshot, crop, heavy JPEG, social-media upload, extreme motion blur, AI upscale, missing metadata.  |

## Final Classification Protocol

1. Gather evidence from at least three independent categories before making a strong AI claim.
2. If evidence is mostly style-based, classify as Real but Heavily Edited or Inconclusive.
3. If evidence is localised to inserted areas, consider Composite/Manipulated instead of fully AI-generated.
4. If provenance exists, report it separately from visual analysis.
5. Use cautious language: likely, consistent with, suggestive of, inconclusive. Avoid absolute claims.



## Gemini Response Templates

| Case                    | Template                                                                                                                                                                                                                   |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Likely real             | This reads as a likely authentic photograph. I see normal photographic/editing traits such as [traits], but I do not see strong structural AI artifacts such as impossible anatomy, pseudo-text, or contradictory shadows. |
| Real but heavily edited | This looks like a real photo with heavy editing. The suspicious traits are mostly Lightroom/editorial artifacts: [traits]. I would not classify it as AI based on those alone.                                             |
| Likely AI-generated     | This is likely AI-generated because multiple structural categories fail: [anatomy/text/shadows/material/geometry]. Confidence: [range]. Caveat: [resolution/compression/provenance limitation].                            |
| Composite/manipulated   | This appears to be a composite or manipulated image. The subject/background integration shows [edge/shadow/noise/perspective] inconsistencies while other areas remain photographically plausible.                         |
| Inconclusive            | The image is too compressed/low-resolution/edited to classify confidently. I can identify visible artifacts, but they are not enough to separate AI generation from editing or compression.                                |

## Quick Reference Summary

- Strong AI evidence = structure, geometry, text, shadows, reflections, impossible object logic.
- Weak AI evidence = smooth skin, contrast, colour grading, vignette, HDR, sharpening, beauty retouching.
- Always separate AI-generation artifacts from Lightroom/editorial artifacts.
- Phone processing and social compression can create plastic skin, smeared hands, halos, noise reduction, and block artifacts.
- When user corrects the model, update the analysis instead of defending the first answer.

## Source Notes and Reliability Labels

This document gives operational image-analysis and image-editing guidance. It must not claim legal or forensic certainty without original files, metadata, provenance, and human review. Use probability language.

- **Internal AuthenticLens Document 1 v1.0:** Source baseline for the AI artifact lexicon: anatomical anomalies, text failures, pattern distortions, boundary failures, AI glow, material physics, structural coherence, and model signatures.

- **Internal AuthenticLens Document 2 v1.0:** Source baseline for photographic physics: optical aberrations, vignetting, distortion, flare, diffraction, sensor noise, film grain, depth of field, reflection physics, exposure, tone, and colour temperature.
- **Internal AuthenticLens Document 3 v1.0:** Source baseline for realism prompting: positive/negative realism vocabulary, film stock and camera references, scenario prompt formulas, and platform strategies.
- **Google AI Developers - Image generation with Gemini:** Current source note: Gemini supports conversational image generation/editing with text, images, or both. Nano Banana refers to Gemini 2.5 Flash Image using model gemini-2.5-flash-image. Nano Banana Pro refers to Gemini 3 Pro Image Preview using model gemini-3-pro-image-preview. Google states generated images include SynthID watermarking.
- **Adobe Lightroom Help - Effects and Details:** Current source note: Texture can smooth or accentuate texture without changing colour/tonality; Clarity changes edge contrast; Dehaze changes haze and can affect perceived detail/colour; Vignette and Grain are stylistic effects; Detail controls Sharpening, Noise Reduction, and Color Noise Reduction.
- **C2PA Technical Specification:** Current source note: Content Credentials use signed manifests and assertions to record provenance and edits. Absence of C2PA data is not proof that an image is fake, and presence of metadata should be interpreted with context.
- **W3C / IEC sRGB references:** Current source note: sRGB is the default web colour space; for e-commerce and social delivery, unmanaged or non-sRGB exports can shift colour appearance across devices.
- **IPTC Photo Metadata Standard:** Current source note: IPTC metadata structures descriptive, location, rights, and administrative data for professional image workflows. Metadata is useful context but is not standalone authenticity proof.

Reliability labels used inside the document: Confirmed source, Common photography practice, Image-forensics heuristic, Operational guidance, Prompting heuristic, Model-behaviour observation.